CS5275 – The Algorithm Designer’s Toolkit (S2 AY2025/26)

Lecture 10:

Boolean functions – Fourier analysis

Boolean functions

- We focus on Boolean functions:

$$f\,:\,\{0,1\}^n \to \{0,1\}$$

- Boolean functions arise in many areas of computer science and beyond:
  - Circuit design
  - Graph theory
  - Combinatorics
  - Coding theory
  - Learning theory
  - Social choice theory
  - …

Boolean functions

Sometimes we allow the output to be a real number.

- We focus on Boolean functions:

$$f : \{0,1\}^n \to \{0,1\}$$

We are flexible about how the bits are represented:

- Sometimes we use **true** and **false**.
- Sometimes we use -1 and 1.

Fourier expansion

- The **Fourier expansion** of a Boolean function $f : \{-1,1\}^n \to \mathbb{R}$ is its representation as a real, <u>multilinear</u> polynomial.

The exponent of every variable in every term is zero or one.

- $x_1 x_2^3 x_3$ is not allowed.
- $x_1 x_2$ is allowed.

Indeed, since $x_i \in \{-1,1\}$, we can replace any x_i^2 with 1.

Fourier expansion

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Example: The maximum function on two bits,

$$\max_2 : \{-1,1\}^2 \to \{-1,1\}.$$

- $\max_2(1,1) = 1$
- $\max_2(-1,1) = 1$
- $\max_2(1,-1) = 1$
- $\max_2(-1,-1) = -1$

Fourier expansion: $\max_2(x_1, x_2) = \frac{1}{2} + \frac{1}{2}x_1 + \frac{1}{2}x_2 - \frac{1}{2}x_1x_2.$

Fourier expansion

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Example: The majority function on three bits,

$$\mathrm{maj}_3 : \{-1,1\}^3 \to \{-1,1\}.$$

Fourier expansion: $\mathrm{maj}_3(x_1, x_2, x_3) = \frac{1}{2}x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_3 - \frac{1}{2}x_1x_2x_3.$

Fourier expansion

- The **Fourier expansion** of a Boolean function $f : \{-1,1\}^n \to \mathbb{R}$ is its representation as a real, multilinear polynomial.

Theorem: For any Boolean function, there exists a unique Fourier expansion.

Fourier expansion: existence

- The **Fourier expansion** of a Boolean function $f : \{-1,1\}^n \to \mathbb{R}$ is its representation as a real, multilinear polynomial.

Theorem: For any Boolean function, there exists a unique Fourier expansion.

For each point $a = (a_1, \dots, a_n) \in \{-1,1\}^n$, consider the **indicator polynomial**:

$$1_{\{a\}}(x) = \left(\frac{1 + a_1 x_1}{2}\right)\left(\frac{1 + a_2 x_2}{2}\right)\cdots\left(\frac{1 + a_n x_n}{2}\right) = \begin{cases} 1, & x = a \\ 0, & x \neq a \end{cases}$$

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Then f has the polynomial representation:

$$f(x) = \sum_{a \in \{-1,1\}^n} f(a) 1_{\{a\}}(x) \longrightarrow$$ **Existence** of a Fourier expansion

Fourier coefficients

The **Fourier expansion** of $f : \{-1,1\}^n \to \mathbb{R}$ can be written as

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S)\chi_S\,,$$

where $\chi_S = x^S = \prod_{i \in S} x_i$ is the **parity function** for S.

The parity of the number of indices $i \in S$ with $x_i = -1$.

Linear algebra

A function $f : \{-1,1\}^n \to \mathbb{R}$ can be viewed as a $\overbrace{2^n\text{-dimensional vector}}^{\text{Listing } f(x) \text{ for all } 2^n \text{ choices of } x \in \{-1,1\}^n}$.

Linear algebra

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$$\mathrm{max}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

Inner product

Listing $f(x)$ for all 2^n choices of $x \in \{-1,1\}^n$

A function $f : \{-1,1\}^n \to \mathbb{R}$ can be viewed as a 2^n-dimensional vector.

The **inner product** is defined as follows:

$$\langle f, g \rangle = 2^{-n} \sum_{x \in \{-1,1\}^n} f(x) g(x)$$

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The 2^{-n} factor ensures that any function $f : \{-1,1\}^n \to \{-1,1\}$ is a unit vector: $\|f\|_2 = \sqrt{\langle f, f \rangle} = 1$.

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$$\langle f, g \rangle = 2^{-n} \sum_{x \in \{-1,1\}^n} f(x) g(x) = \mathbb{E}_{\boldsymbol{x} \sim \{-1,1\}^n}[f(\boldsymbol{x}) g(\boldsymbol{x})]$$

Notation:

- $\boldsymbol{x} \sim \{-1,1\}^n$ means that $\boldsymbol{x}$ is chosen uniformly at random from $\{-1,1\}^n$.
- We write random variables in **boldface**.
- Unless otherwise specified, we consider the uniform random distribution $\boldsymbol{x} \sim \{-1,1\}^n$.
- Thus, we may write $\mathbb{E}_{\boldsymbol{x} \sim \{-1,1\}^n}[f(\boldsymbol{x}) g(\boldsymbol{x})] = \mathbb{E}_{\boldsymbol{x}}[f(\boldsymbol{x}) g(\boldsymbol{x})] = \mathbb{E}[f(\boldsymbol{x}) g(\boldsymbol{x})] = \mathbb{E}[fg]$.

Inner product

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More generally, we define $\|f\|_p = \left(\mathbb{E}_{\boldsymbol{x} \sim \{-1,1\}^n}[|f(\boldsymbol{x})|^p]\right)^{1/p}$.

Fourier expansion: uniqueness

- **Claim:** The collection of 2^n parity functions is an orthonormal basis.

The Fourier expansion $f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S$ is **unique**.

Fourier expansion: uniqueness

- **Claim:** The collection of 2^n parity functions is an orthonormal basis.
- To prove the claim, it suffices to show:

$$\langle \chi_S, \chi_T \rangle = \begin{cases} 1, & S = T \\ 0, & S \neq T \end{cases}$$

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$$\langle \chi_S, \chi_T \rangle = \begin{cases} 1, & S = T \\ 0, & S \neq T \end{cases}$$

Observation: $\mathbb{E}[\chi_S(\boldsymbol{x})] = \mathbb{E}[\prod_{i \in S} \boldsymbol{x}_i] = \prod_{i \in S} \underbrace{\mathbb{E}[\boldsymbol{x}_i]}_{\mathbb{E}[\boldsymbol{x}_i] = 0} = \begin{cases} 1, & S = \emptyset \\ 0, & S \neq \emptyset \end{cases}$

Fourier expansion: uniqueness

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$$\langle \chi_S, \chi_T \rangle = \mathbb{E}[\chi_S \chi_T] = \mathbb{E}\big[\chi_{\underbrace{(S\cup T)\setminus(S\cap T)}_{\substack{(S \cup T) \setminus (S \cap T) = \emptyset \\ \text{if and only if } S = T}}}\big]$$

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Basic formulas

Observation: $\hat{f}(S) = \langle f, \chi_S \rangle = \mathbb{E}[f(\boldsymbol{x})\chi_S(\boldsymbol{x})]$

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The Fourier expansion of f.

$$\langle f, \chi_S \rangle = \left\langle \underbrace{\sum_{T \subseteq [n]} \hat{f}(T)\chi_T}_{\text{The Fourier expansion of } f.}, \chi_S \right\rangle = \sum_{T \subseteq [n]} \hat{f}(T)\langle \chi_T, \chi_S \rangle = \hat{f}(S)$$

The inner product is linear in both of its arguments.

$$\langle \chi_T, \chi_S \rangle = \begin{cases} 1, & S = T \\ 0, & S \neq T \end{cases}$$

Basic formulas

Observation: $\hat{f}(S) = \langle f, \chi_S \rangle = \mathbb{E}[f(\boldsymbol{x})\chi_S(\boldsymbol{x})]$

Parseval's theorem: $\|f\|_2^2 = \langle f, f \rangle = \mathbb{E}[f(\boldsymbol{x})^2] = \sum_{S \subseteq [n]} \hat{f}(S)^2$

Intuition: Distance is preserved under a change of orthonormal basis.

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$\uparrow f = g$

Plancherel's theorem: $\langle f, g \rangle = \mathbb{E}[fg] = \sum_{S\subseteq[n]} \hat{f}(S)\hat{g}(S)$

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Plancherel's theorem: $\langle f, g \rangle = \mathbb{E}[fg] = \sum_{S \subseteq [n]} \hat{f}(S)\hat{g}(S)$

$\uparrow$

$$\langle f, g \rangle = \left\langle \sum_{S \subseteq [n]} \hat{f}(S)\chi_S, \sum_{T \subseteq [n]} \hat{g}(T)\chi_T \right\rangle = \sum_{S, T \subseteq [n]} \hat{f}(S)\hat{g}(T)\langle \chi_S, \chi_T \rangle = \sum_{S \subseteq [n]} \hat{f}(S)\hat{g}(S)$$

Fourier expansion

Linearity of $\langle \cdot, \cdot \rangle$

$\langle \chi_S, \chi_T \rangle = \begin{cases} 1, & S = T \\ 0, & S \neq T \end{cases}$

Basic formulas

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Corollary: For any Boolean function $f : \{-1,1\}^n \to \{-1,1\}$,

$$\sum_{S\subseteq[n]} \hat{f}(S)^2 = 1$$

The sum of **Fourier weights** $\hat{f}(S)^2$ equals 1.

Information captured by Fourier coefficients

- A basic theme in the analysis of Boolean functions:
  - Interesting properties of f are captured by its Fourier coefficients.

Observation: $\mathbb{E}[f] = \hat{f}(\emptyset)$

$$\mathbb{E}[f] = \mathbb{E}[f(\boldsymbol{x})\chi_{\emptyset}(\boldsymbol{x})] = \langle f, \chi_{\emptyset} \rangle = \hat{f}(\emptyset)$$

$$\chi_{\emptyset}(\boldsymbol{x}) = 1$$

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Observation: $\mathrm{Var}[f] = \sum_{S \subseteq [n] \setminus \emptyset} \hat{f}(S)^2$

$$\mathrm{Var}[f] = \mathbb{E}[f(\boldsymbol{x})^2] - \mathbb{E}[f(\boldsymbol{x})]^2 = \left(\sum_{S \subseteq [n]} \hat{f}(S)^2 \right) - \hat{f}(\emptyset)^2$$

Parseval's theorem: $\mathbb{E}[f(\boldsymbol{x})^2] = \sum_{S \subseteq [n]} \hat{f}(S)^2$

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Intuition: Fourier analysis is often useful when a problem involves <u>counting</u> or <u>uniform distribution</u>.

Outlook

- **Next:** An application of Fourier analysis of Boolean functions.

Given query access to an unknown function $f : \mathbb{F}_2^n \to \mathbb{F}_2$,
how can we efficiently distinguish between the two cases?

- f is linear.
- f is far from linear.

References

- Sections 1.1–1.4 of https://arxiv.org/abs/2105.10386